

Assignment 1

Computer Programming with C

MCA102

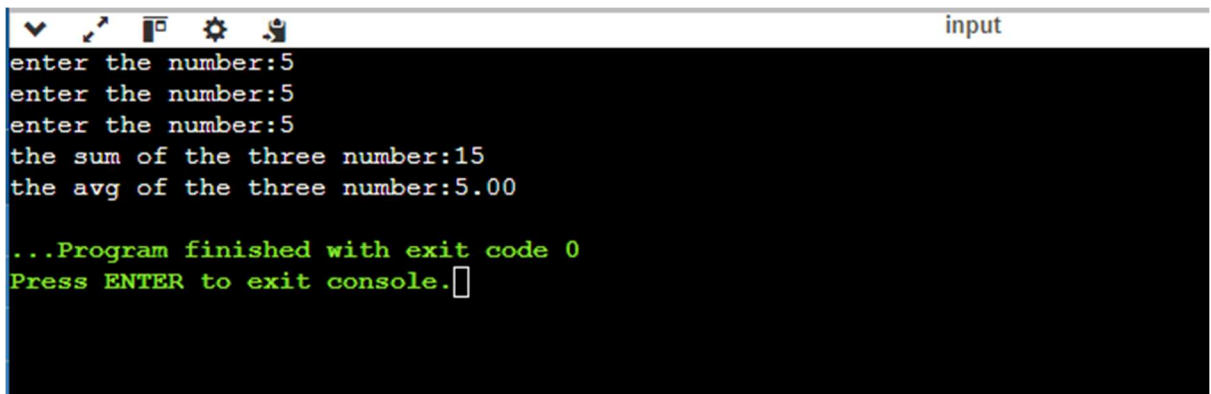
Topic: C Basics and Operators

- 1) Write a C program to find the sum and average of three numbers.

Input:-

```
#include<stdio.h>
int main()
{
    int a1,a2,a3,sum=0;
    float avg=0.0;
    printf("enter the number:");
    scanf("%d",&a1);
    printf("enter the number:");
    scanf("%d",&a2);
    printf("enter the number:");
    scanf("%d",&a3);
    sum+=a1+a2+a3;
    avg=sum/3;
    printf("the sum of the three number:%d\n",sum);
    printf("the avg of the three number:%.2f",avg);
}
```

Output:-



```
input
enter the number:5
enter the number:5
enter the number:5
the sum of the three number:15
the avg of the three number:5.00

...Program finished with exit code 0
Press ENTER to exit console.
```

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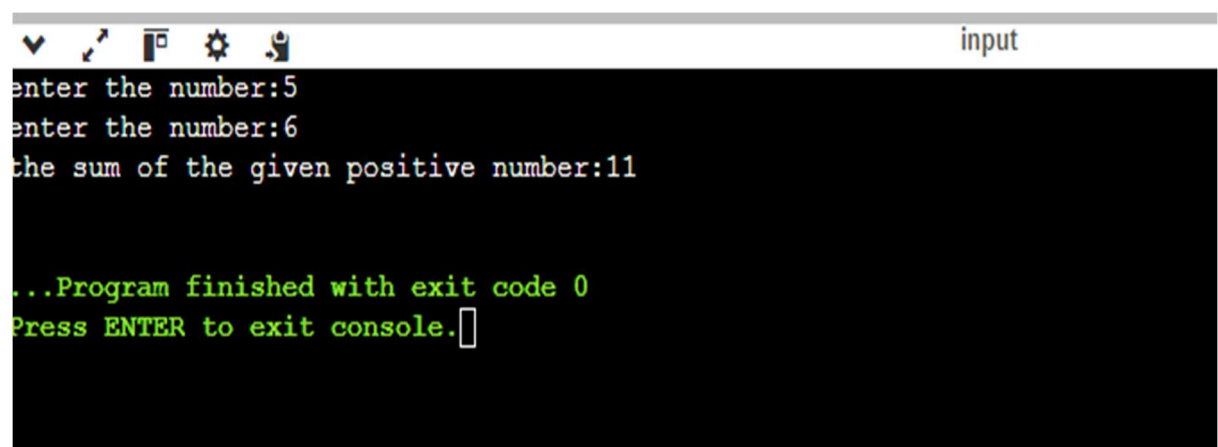
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- 2) Write a C program to find the sum of individual digits of a given positive integer.

Input:-

```
#include<stdio.h>
int main()
{
    int a1,a2,sum=0;
    printf("enter the number:");
    scanf("%d",&a1);
    printf("enter the number:");
    scanf("%d",&a2);
    sum+=a1+a2;
    printf("the sum of the given positive number:%d\n",sum);
}
```

Output:-



```
input
enter the number:5
enter the number:6
the sum of the given positive number:11

...Program finished with exit code 0
Press ENTER to exit console.
```

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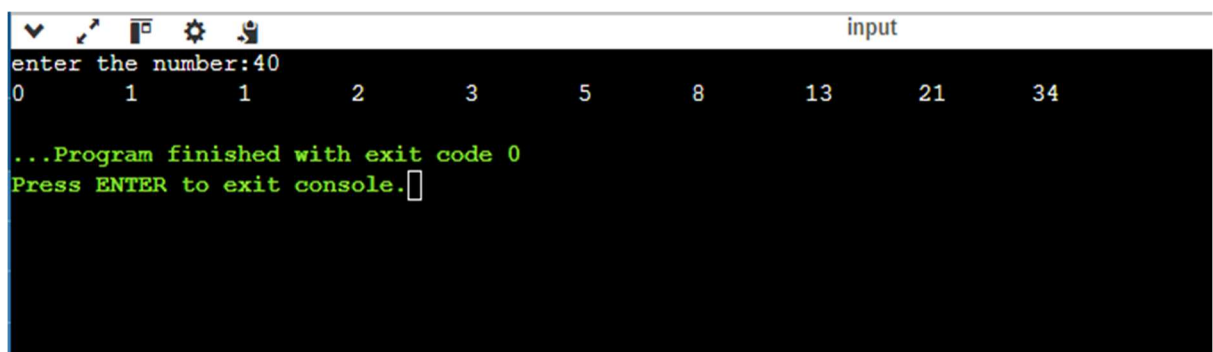
Topic: C Basics and Operators

- 3) Write a C program to generate the first n terms of the Fibonacci sequence.

Input:-

```
#include<stdio.h>
int main()
{
    int n,a=1,b=0,sum=0;
    printf("enter the number:");
    scanf("%d",&n);
    while(sum<=n)
    {
        printf("%d\t",sum);
        sum=a+b;
        a=b;
        b=sum;
    }
}
```

Output:-



The screenshot shows a terminal window with a title bar containing icons for window management and a tab labeled 'input'. The terminal output is as follows:

```
enter the number:40
0      1      1      2      3      5      8      13     21     34

...Program finished with exit code 0
Press ENTER to exit console.
```

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4) Write a C program to generate prime numbers between 1 to n

Input:-

```
#include<stdio.h>
int main()
{
    int n,i,y,count;
    printf("enter the number:");
    scanf("%d",&n);
    printf("prime numbers between 1 to n\n");
    for(i=1;i<=n;i++)
    {
        count=0;
        for(y=1;y<=i;y++)
        {
            if(i%y==0)
            {
                count++;
            }
        }
        if(count==2)
        {
            printf("%d\t",i);
        }
    }
}
```

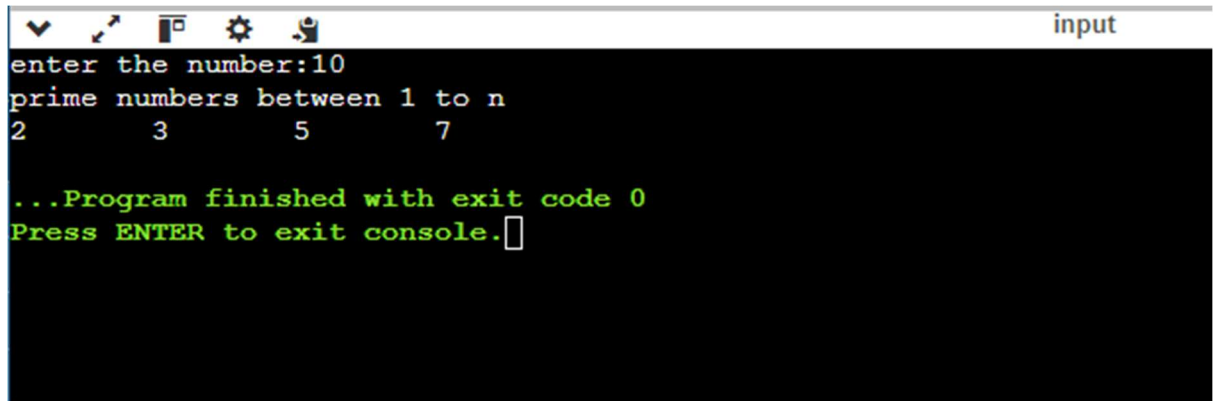
Output:-

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```
input
enter the number:10
prime numbers between 1 to n
2      3      5      7

...Program finished with exit code 0
Press ENTER to exit console.
```

- 5) Write a C program to check whether a given number is an Armstrong number or not.

Input:-

```
#include<stdio.h>
int main()
{
    int n,m,rem,sum=0;
    printf("enter the number:");
    scanf("%d",&n);
    m=n;
    while(n>0)
    {
        rem=n%10;
        sum+=rem*rem*rem;
        n=n/10;
    }
    if(sum==m)
    {
        printf("this number is armstrong number:%d",m);
    }
}
```

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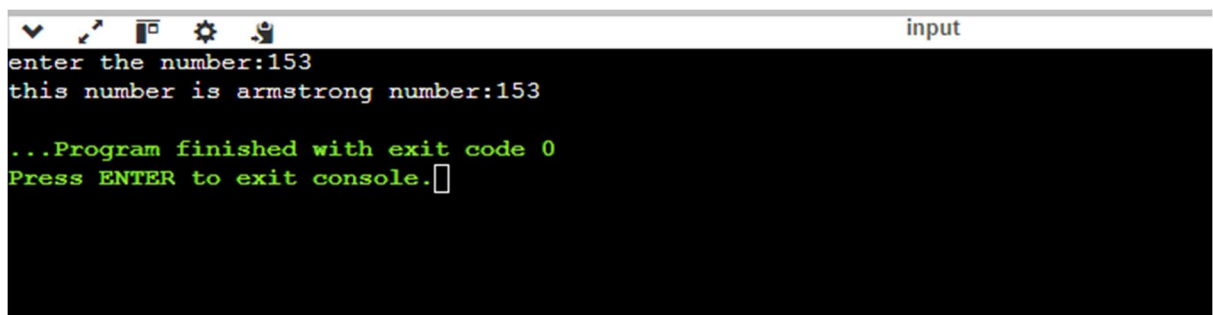
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```
}  
else  
{  
    printf("this number is not armstrong number:%d",m);  
}  
}
```

Output:-



```
input  
enter the number:153  
this number is armstrong number:153  
...Program finished with exit code 0  
Press ENTER to exit console.█
```

- 6) Write a C program to evaluate the algebraic expression $(ax+b)/(ax-b)$.

Input:-

```
#include<stdio.h>  
int main()  
{  
    int a,x,b;  
    float result;  
    printf("enter the number:");  
    scanf("%d",&a);  
    printf("enter the number:");  
    scanf("%d",&x);  
    printf("enter the number:");  
    scanf("%d",&b);
```

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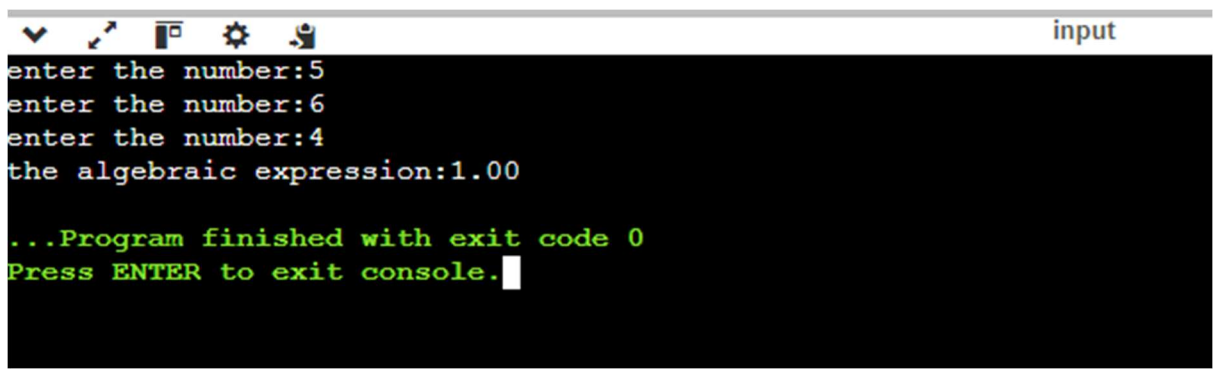
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```
result=(a*x+b)/(a*x-b);
printf("the algebraic expression:%.2f",result);
}
```

Output:-



```
input
enter the number:5
enter the number:6
enter the number:4
the algebraic expression:1.00
...Program finished with exit code 0
Press ENTER to exit console.
```

7) Write a C program to check if the given number is a perfect number.

Input:-

```
#include<stdio.h>
int main()
{
    int a,rem,i,sum=0;
    printf("enter the number:");
    scanf("%d",&a);
    for(i=1;i<a;i++)
    {
        rem=a%i;
        if(rem==0)
        {
            sum+=i;
        }
    }
}
```

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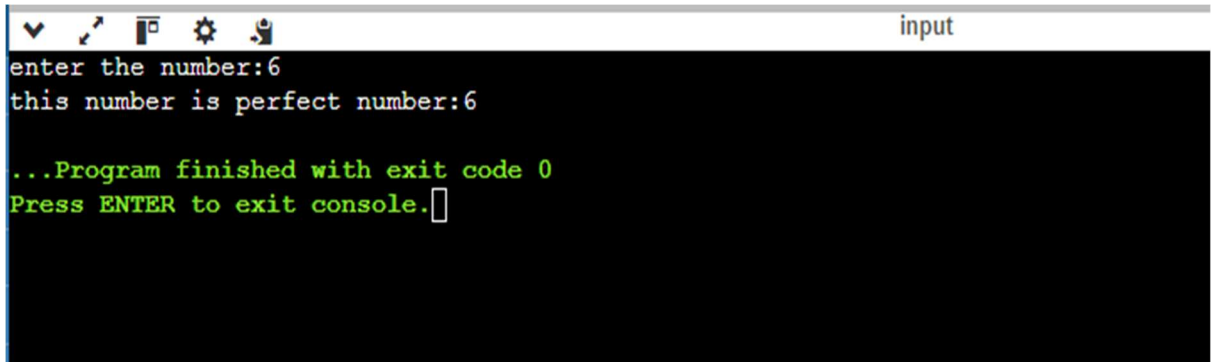
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```
    }  
}  
if(sum==a)  
{  
    printf("this number is perfect number:%d",a);  
}  
else  
{  
    printf("this number is not perfect number:%d",a);  
}  
}
```

Output:-



```
input  
enter the number:6  
this number is perfect number:6  
  
...Program finished with exit code 0  
Press ENTER to exit console.
```

- 8) Write a C program to check if a given number is a strong number.

Input:-

```
#include<stdio.h>  
int main()  
{  
    int a,rem,i,sum=0,temp,fact;  
    printf("enter the number:");
```


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```
scanf("%d",&a);
temp=a;
while(a)
{
    i=1,fact=1;
    rem=a%10;
    while(i<=rem)
    {
        fact=fact*i;
        i++;
    }
    sum=sum+fact;
    a=a/10;
}
if(sum==temp)
{
    printf("this number is strong number:%d",temp);
}
else
{
    printf("this number is not strong number:%d",temp);
}
}
```

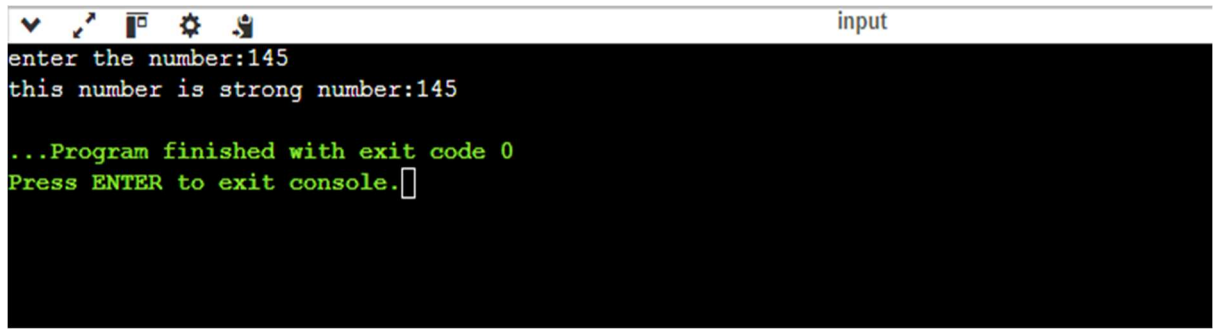
Output:-

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```
input
enter the number:145
this number is strong number:145

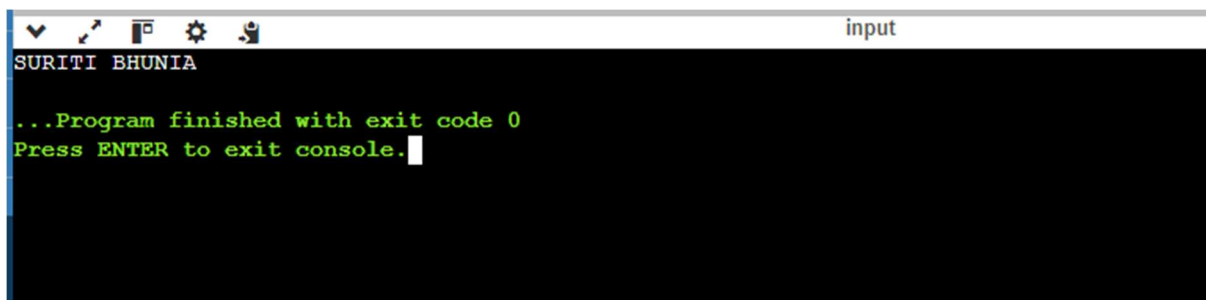
...Program finished with exit code 0
Press ENTER to exit console.
```

- 9) Write a program to print your name without using any semicolons in the program.

Input:-

```
#include<stdio.h>
int main()
{
    if(printf("SURITI BHUNIA")){}
}
```

Output:-



```
input
SURITI BHUNIA

...Program finished with exit code 0
Press ENTER to exit console.
```

- 10) Write a program to convert temperatures in Celsius to Fahrenheit and vice-versa.

Input:-

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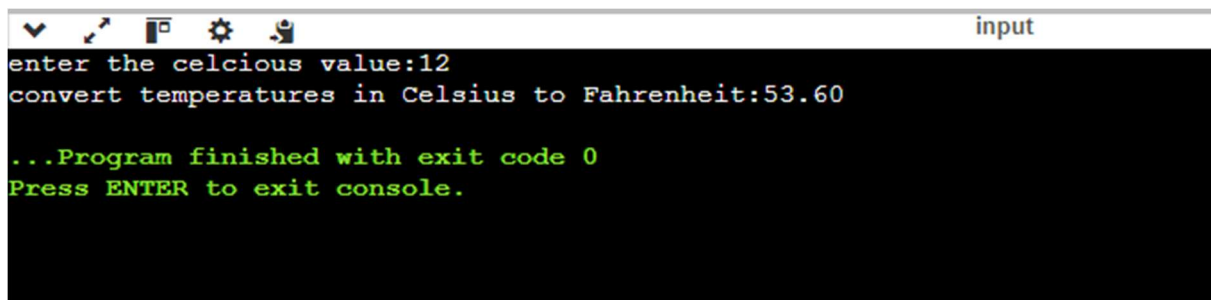
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```
#include<stdio.h>
int main()
{
    float cel,fahr;
    printf("enter the celcius value:");
    scanf("%f",&cel);
    fahr=(cel*9/5)+32;
    printf("convert temperatures in Celsius to
Fahrenheit:%.2f",fahr);
}
```

Output:-



```
input
enter the celcius value:12
convert temperatures in Celsius to Fahrenheit:53.60
...Program finished with exit code 0
Press ENTER to exit console.
```

- 11) Write a C program to check whether a number is a palindrome or not.

Input:-

```
#include<stdio.h>
int main()
{
    int n,rem,m,rev=0;
    printf("enter the number:");
    scanf("%d",&n);
    m=n;
```

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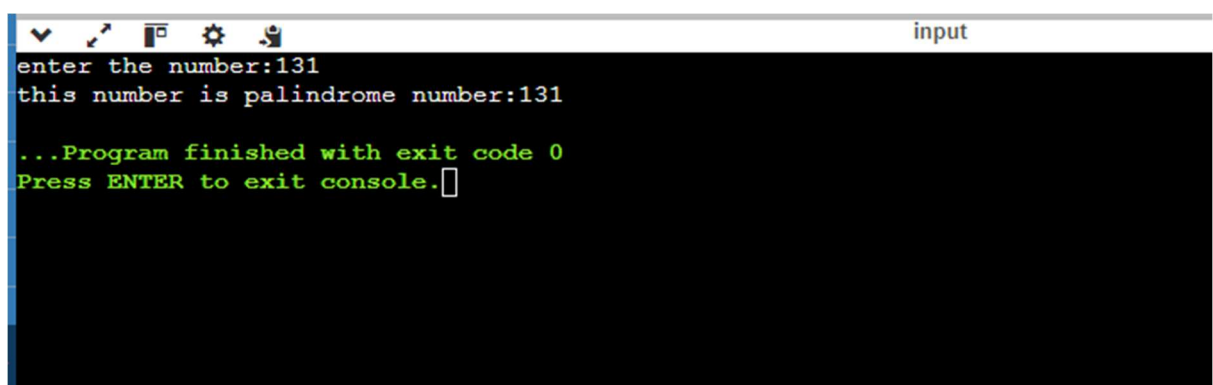
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```
while(n>0)
{
    rem=n%10;
    rev=rev*10+rem;
    n=n/10;
}
if(rev==m)
{
    printf("this number is palindrome number:%d",m);
}
else
{
    printf("this number is not palindrome number:%d",m);
}
}
```

Output:-



```
input
enter the number:131
this number is palindrome number:131
...Program finished with exit code 0
Press ENTER to exit console.
```

- 12) Write a C program to find the maximum between two numbers.

Input:-

```
#include<stdio.h>
```

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```
int main()
{
    int n,m;
    printf("enter the number:");
    scanf("%d",&n);
    printf("enter the number:");
    scanf("%d",&m);
    if(n>m)
    {
        printf("1st number is maximum:%d",n);
    }
    else
    {
        printf("2nd number is maximum:%d",m);
    }
}
```

Output:-



```
input
enter the number:6
enter the number:5
1st number is maximum:6

...Program finished with exit code 0
Press ENTER to exit console.
```

- 13) Write a C program to find the maximum between three numbers.

Input:-

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```
#include<stdio.h>
int main()
{
    int a,b,c;
    printf("enter the number:");
    scanf("%d",&a);
    printf("enter the number:");
    scanf("%d",&b);
    printf("enter the number:");
    scanf("%d",&c);
    if(a>b)
    {
        if(a>c)
        {
            printf("1st number is maximum:%d",a);
        }
        else
        {
            printf("3rd number is maximum:%d",c);
        }
    }
    else
    {
        if(b>c)
        {
            printf("2nd number is maximum:%d",b);
        }
        else
        {
            printf("3rd number is maximum:%d",c);
        }
    }
}
```

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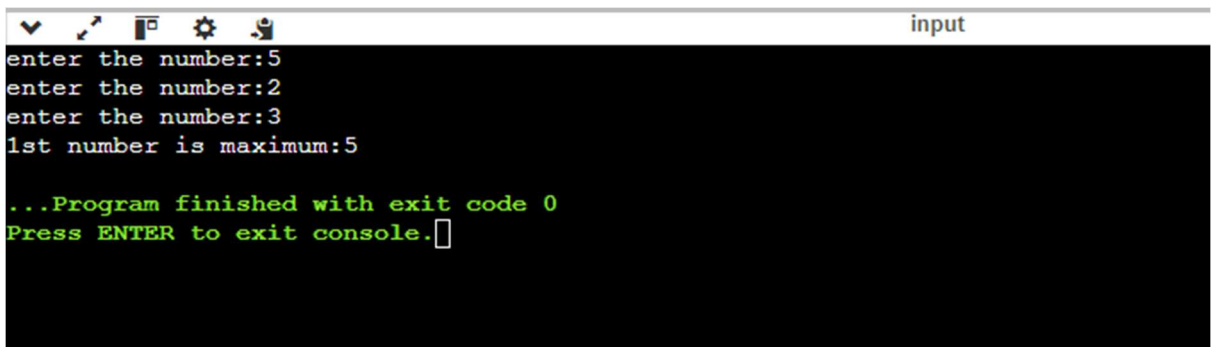
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```
}  
}  
}
```

Output:-



```
input  
enter the number:5  
enter the number:2  
enter the number:3  
1st number is maximum:5  
...Program finished with exit code 0  
Press ENTER to exit console.
```

- 14) Write a C program to check whether a number is negative, positive, or zero.

Input:-

```
#include<stdio.h>  
int main()  
{  
    int a;  
    printf("enter the number:");  
    scanf("%d",&a);  
    if(a>0)  
    {  
        printf("this number is positive number:%d",a);  
    }  
    else if(a<0)  
    {  
        printf("this number is negetive number:%d",a);  
    }  
}
```

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```
}  
else  
{  
    printf("this number is zero number:%d",a);  
}  
}
```

Output:-

```
input  
enter the number:-5  
this number is negative number:-5  
  
...Program finished with exit code 0  
Press ENTER to exit console.
```

- 15) Write a C program to check whether a number is divisible by 5 and 11 or not within the range of 100 to 500.

Input:-

```
#include<stdio.h>  
int main()  
{  
    int a,b,i;  
    printf("enter the starting number:");  
    scanf("%d",&a);  
    printf("enter the ending number:");  
    scanf("%d",&b);
```


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```
printf("\na number is divisible by 5 and 11 or not within the
range of 100 to 500:\n");
for(i=a;i<=b;i++)
{
    if((i%5==0) || (i%11==0))
    {
        printf("%d\t",i);
    }
}
```

Output:-

```
input
enter the starting number:100
enter the ending number:500
a number is divisible by 5 and 11 or not within the range of 100 to 500:
100 105 110 115 120 121 125 130 132 135 140 143 145 150 154 1
55 160 165 170 175 176 180 185 187 190 195 198 200 205 209 2
10 215 220 225 230 231 235 240 242 245 250 253 255 260 264 2
65 270 275 280 285 286 290 295 297 300 305 308 310 315 319 3
20 325 330 335 340 341 345 350 352 355 360 363 365 370 374 3
75 380 385 390 395 396 400 405 407 410 415 418 420 425 429 4
30 435 440 445 450 451 455 460 462 465 470 473 475 480 484 4
85 490 495 500
...Program finished with exit code 0
```

16) Write a C program to check whether a number is even or odd.

Input:-

```
#include<stdio.h>
int main()
{
    int a;
    printf("enter the starting number:");
```

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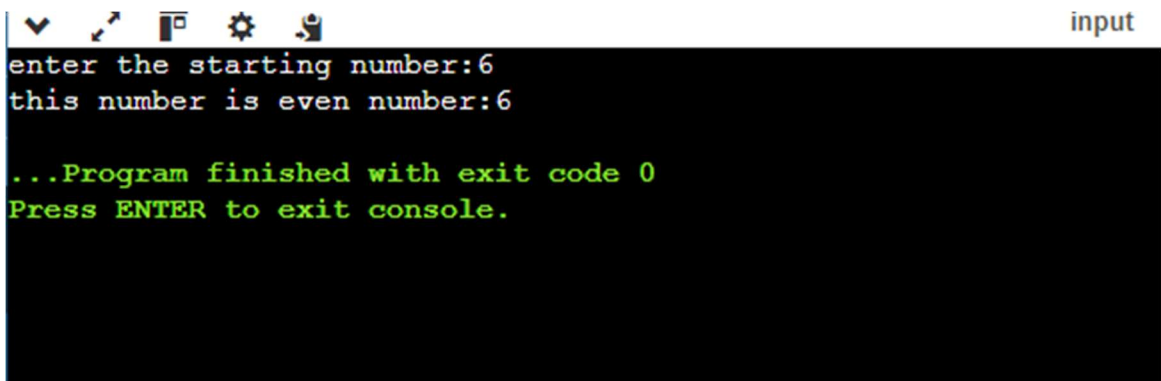
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```
scanf("%d",&a);
if(a%2==0)
{
    printf("this number is even number:%d",a);
}
else
{
    printf("this number is odd number:%d",a);
}
}
```

Output:-



```
enter the starting number:6
this number is even number:6

...Program finished with exit code 0
Press ENTER to exit console.
```

- 17) Write a C program to check whether a year is a leap year or not.

Input:-

```
#include<stdio.h>
int main()
{
    int a;
    printf("enter the year:");
```

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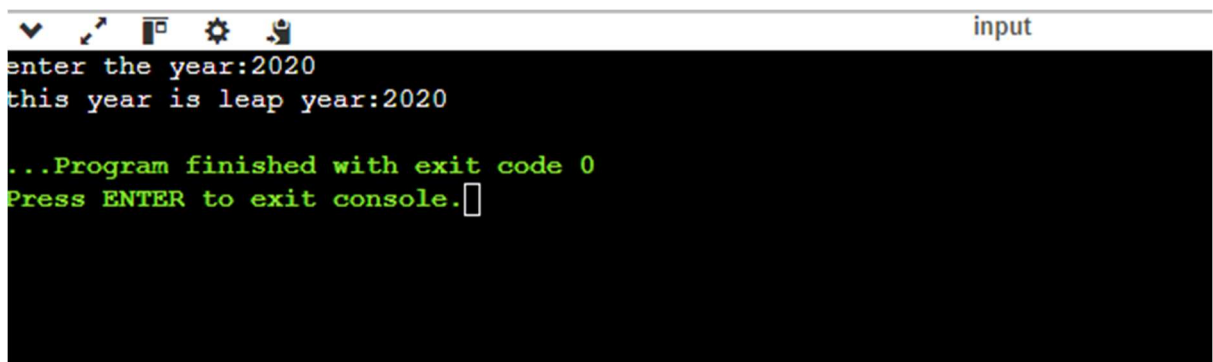
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```
scanf("%d",&a);
if(a%100==0)
{
    printf("this year is leap year:%d",a);
}
else if(a%400==0)
{
    printf("this year is not leap yea:%d",a);
}
else if(a%4==0)
{
    printf("this year is leap year:%d",a);
}
else
{
    printf("this year is not leap year:%d",a);
}
}
```

Output:-

A screenshot of a terminal window with a black background and white text. The window title is 'input'. The text shows the program execution: 'enter the year:2020', 'this year is leap year:2020', and a green message '...Program finished with exit code 0'. It also shows 'Press ENTER to exit console.' with a cursor. The terminal has standard icons in the top-left corner.

```
input
enter the year:2020
this year is leap year:2020
...Program finished with exit code 0
Press ENTER to exit console.
```

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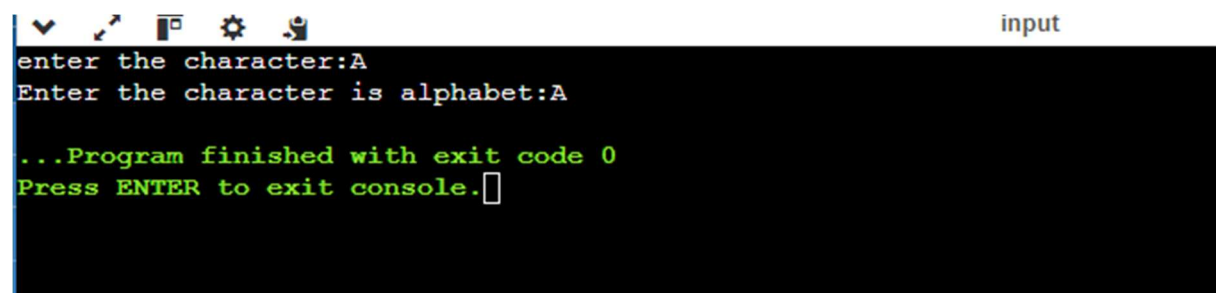
Topic: C Basics and Operators

18) Write a C program to check whether a character is alphabet or not.

Input:-

```
#include<stdio.h>
#include<string.h>
int main()
{
    char a;
    printf("enter the character:");
    scanf("%c",&a);
    if((a>='A' && a<='Z') || (a>='a' && a<='z'))
    {
        printf("Enter the character is alphabet:%c",a);
    }
    else
    {
        printf("Enter the character is not alphabet:%c",a);
    }
}
```

Output:-

A screenshot of a terminal window titled 'input'. The terminal shows the execution of a C program. The first prompt is 'enter the character:A', where the user has entered 'A'. The second prompt is 'Enter the character is alphabet:A', where the user has entered 'A'. The program then displays '...Program finished with exit code 0' and 'Press ENTER to exit console.' with a cursor. The terminal has a dark background with light-colored text.

```
input
enter the character:A
Enter the character is alphabet:A
...Program finished with exit code 0
Press ENTER to exit console.
```

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- 19) Write a C program to input any alphabet and check whether it is a vowel or consonant.

Input:-

```
#include<stdio.h>
#include<string.h>
int main()
{
    char a;
    printf("enter the character:");
    scanf("%c",&a);
    if((a=='a' || a=='e' || a=='i' || a=='o' || a=='u')
    || (a=='A' || a=='E' || a=='I' || a=='O' || a=='U'))
    {
        printf("Enter the character is vowel:%c",a);
    }
    else
    {
        printf("Enter the character is consonant:%c",a);
    }
}
```

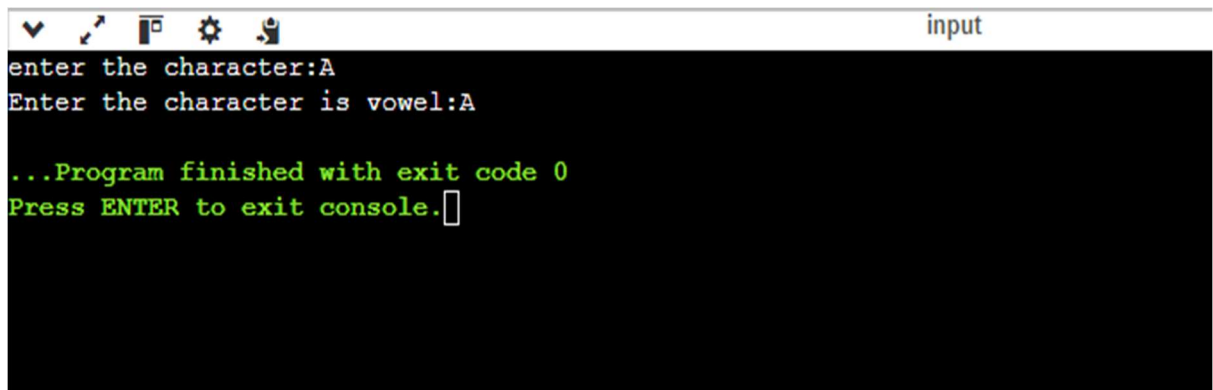
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```
input
enter the character:A
Enter the character is vowel:A
...Program finished with exit code 0
Press ENTER to exit console.
```

- 20) Write a C program to input any character and check whether it is an alphabet, digit, or special character

Input:-

```
#include<stdio.h>
#include<string.h>
int main()
{
    char a;
    printf("enter the character:");
    scanf("%c",&a);
    if((a>='A' && a<='Z') || (a>='a' && a<='z'))
    {
        printf("Enter the character is alphabet:%c",a);
    }
    else if(a>='0' && a<='9')
    {
```

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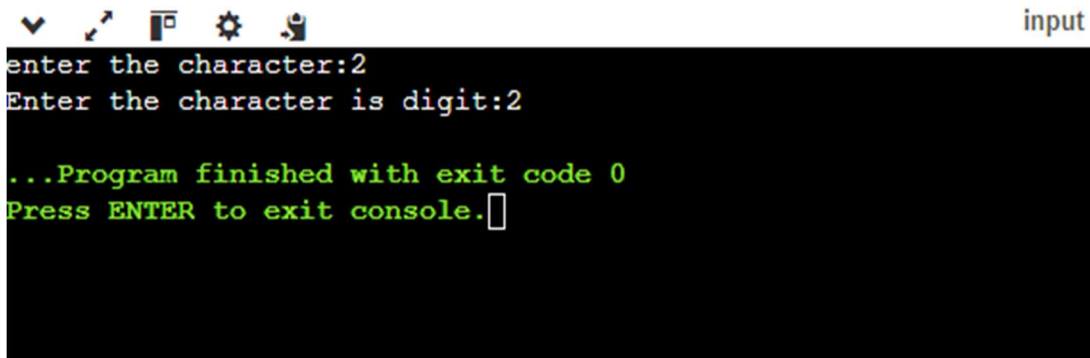
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```
printf("Enter the character is digit:%c",a);  
}  
else  
{  
    printf("Enter the character is special character:%c",a);  
}  
}
```

Output:-



```
enter the character:2  
Enter the character is digit:2  
  
...Program finished with exit code 0  
Press ENTER to exit console.
```